Important Note: If $\overline{X}$, $\overline{Y}$ are not integers then the above method is tedious and time consuming to calculate b_{YX} and b_{XY}. The following modified formulae are easy for calculation.

$$b_{YX} = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}$$

$$b_{XY} = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} y_i^2 - \left(\sum_{i=1}^{n} y_i\right)^2}$$

Example 5.4

The following data gives the experience of machine operators and their performance ratings as given by the number of good parts turned out per 50 pieces.

Operators	1	2	3	4	5	6	7	8
Experience (X)	8	11	7	10	12	5	4	6
Ratings (Y)	11	30	25	44	38	25	20	27

Obtain the regression equations and estimate the ratings corresponding to the experience x=15.

Solution:

	x_i	y_i	$x_i y_i$	x_i^2	y_i^2
	8	11	88	64	121
	11	30	330	121	900
	7	25	175	49	625
	10	44	440	100	1936
	12	38	456	144	1444
	5	25	125	25	625
	4	20	80	16	400
	6	27	162	36	729
Total	63	220	1856	555	6780

Regression equation of Y on X,

$$\hat{Y} - \overline{y} = b_{YX}\left(x - \overline{x}\right)$$

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{63}{8} = 7.875$$

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{220}{8} = 27.5$$

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